

# Limits And Derivatives :

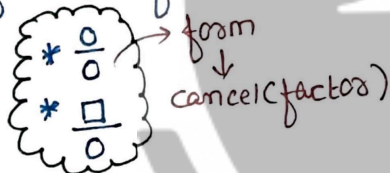
What is a Limit?

A limit tells us what value a function is approaching as  $x$  gets closer to a number.

| Formula                                         | Result        |
|-------------------------------------------------|---------------|
| $\lim_{x \rightarrow 0} \frac{\sin x}{x}$       | 1             |
| $\lim_{x \rightarrow 0} \frac{\tan x}{x}$       | 1             |
| $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x}$   | 0             |
| $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}$ | $\frac{1}{2}$ |
| $\lim_{x \rightarrow 0} \frac{\sin ax}{x}$      | a             |
| $\lim_{x \rightarrow 0} \frac{\tan ax}{x}$      | a             |

What to do when 0/0 and a/0 form comes in limits?

In limits 0/0 and a/0 form develop then you can put limit into the function rather simplify function first or use formula.



Algebra of Limit -

In simple words:

- Limit work just like normal algebra. You can add, subtract, multiply or divide limits - Provided the individual limits exist and denominator is zero.

Suppose  $f(x)$  and  $g(x)$  are two functions, and their limits as  $2a$  exist. Then

1. Sum Rule

$$\lim_{x \rightarrow a} [f(x) + g(x)] = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x)$$

2. Difference Rule

$$\lim_{x \rightarrow a} [f(x) - g(x)] = \lim_{x \rightarrow a} f(x) - \lim_{x \rightarrow a} g(x)$$

3. Product Rule

$$\lim_{x \rightarrow a} [f(x) \cdot g(x)] = \left( \lim_{x \rightarrow a} f(x) \right) \cdot \left( \lim_{x \rightarrow a} g(x) \right)$$

4. Quotient Rule (When denominator limit  $\neq 0$ )

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)}$$

Special Case:

If  $g(x) = \lambda$  (a constant), then

$$\lim_{x \rightarrow a} \lambda \cdot f(x) = \lambda \cdot \lim_{x \rightarrow a} f(x)$$

Example:

Function:  $f(x) = x^2$  As  $x \rightarrow 0$

$$f(x) = x^2 \rightarrow 0$$

$$\text{So, } \lim_{x \rightarrow 0} f(x) = 0$$

$$\lim_{x \rightarrow 3} (x+3)$$

$$\lim_{x \rightarrow \pi} \left( x + \frac{22}{7} \right)$$

Soln: Applying Limit

$$\Rightarrow 3+3$$

$$\Rightarrow \boxed{6} \text{ Ans.}$$

Soln: By Applying Limit

$$\Rightarrow \left( \pi + \frac{22}{7} \right)$$

$$\Rightarrow \left( \frac{22}{7} + \frac{22}{7} \right) \Rightarrow \boxed{\frac{44}{7}} \text{ Ans.}$$

Example:

$$\lim_{x \rightarrow 4} \frac{4x+3}{x-2} \Rightarrow \frac{4(4)+3}{4-2} \Rightarrow \frac{16+3}{2} \Rightarrow \boxed{\frac{19}{2}}$$

$$\lim_{x \rightarrow 2} \frac{3x^2-x-10}{x^2-4} \Rightarrow \frac{3(2)^2-2-10}{2^2-4} \Rightarrow \frac{12-12}{4-4} = \frac{0}{0}$$

Factorization Method.

$$= \lim_{x \rightarrow 2} \frac{3x^2-6x+5x-10}{x^2-2^2}$$

$$= \lim_{x \rightarrow 2} \frac{3x(x-2)+5(x-2)}{(x+2)(x-2)}$$

$$= \lim_{x \rightarrow 2} \frac{(x-2)(3x+5)}{(x+2)(x-2)} \Rightarrow \lim_{x \rightarrow 2} \frac{3x+5}{x+2}$$

By Applying Limit

$$\Rightarrow \frac{3(2)+5}{2+2} \Rightarrow \frac{6+5}{4} \Rightarrow \left(\frac{11}{4}\right) \text{ Ans.}$$

**Question)** Find the limits:  $\lim_{x \rightarrow 1} [x^3 - x^2 + 1]$

$$\Rightarrow 1^3 - 1^2 + 1 \Rightarrow 1 - 1 + 1 \Rightarrow (1) \text{ Ans.}$$

**Question)** Find the limits:

$$\lim_{x \rightarrow -1} [1 + x + x^2 + \dots + x^{10}]$$

$$\Rightarrow \lim_{x \rightarrow -1} 1 + x + x^2 + x^3 + x^4 + x^5 + x^6 + x^7 + x^8 + x^9 + x^{10}$$

$$\Rightarrow 1 + (-1) + (-1)^2 + (-1)^3 + (-1)^4 + (-1)^5 + (-1)^6 + (-1)^7 + (-1)^8 + (-1)^9 + (-1)^{10}$$

$$\Rightarrow 1 - 1 + 1 - 1 + 1 - 1 + 1 - 1 + 1 - 1 + 1 - 1 + 1$$

$$\Rightarrow (1) \text{ Ans.}$$

Limits ke Question ko approach kaise karen?? (KBDT)

Polynomials  $\rightarrow$  direct substitution

Rational function  $\rightarrow$

\* If denominator  $\neq 0 \rightarrow$  direct substi

\* If denominator = 0 and numerator  $\neq 0 \rightarrow$  Limit doesn't exist

\* If denominator = 0 and numerator = 0  $\rightarrow$  cancel & Retry.

**\*\*\* Quest)** Formula  $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$

$$\lim_{x \rightarrow 0} \frac{\sin ax}{bx}$$

$$\lim_{x \rightarrow 0} \frac{\sin ax}{\sin bx}, a, b \neq 0$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{\sin ax}{ax} \times \frac{1}{bx} \times ax \Rightarrow \lim_{x \rightarrow 0} \frac{\sin ax}{\sin bx} \times \frac{1}{bx}$$

[M/D by ax]

[M/D by ax]

$$\Rightarrow \lim_{x \rightarrow 0} 1 \times \frac{1}{bx} \times ax \Rightarrow \lim_{x \rightarrow 0} \frac{\sin ax}{ax} \times ax \times \frac{1}{\sin bx}$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{a}{b} \quad [M/D by bx]$$

$$\Rightarrow \left(\frac{a}{b}\right) \text{ Ans.} \quad \Rightarrow \lim_{x \rightarrow 0} \frac{\sin ax}{ax} \times ax \times \frac{bx}{\sin bx} \times \frac{1}{bx}$$

$$\Rightarrow \left(\frac{a}{b}\right) \text{ Ans.}$$

$$\lim_{x \rightarrow 0} \frac{\cos x}{\pi - x}$$

$$\Rightarrow \frac{\cos 0^\circ}{\pi - 0}$$

$$\Rightarrow \frac{1}{\pi}$$

$$\Rightarrow \frac{1 \times 7}{22} \Rightarrow \left(\frac{7}{22}\right) \text{ Ans.}$$

$$\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = n \cdot a^{n-1}$$

- n is any positive integer.
- This comes from derivatives of  $x^n$ .
- Used when direct substitution gives 0/0

**Example:**

$$\lim_{x \rightarrow 3} \frac{x^4 - 81}{x - 3} = \lim_{x \rightarrow 3} \frac{x^4 \cdot 3^4}{x - 3} = 4 \cdot 3^4 = 108$$

**Theorem - 2 (Simple version)**

$$\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = n a^{n-1}, \text{ for any positive integer } n$$

In simple words:

This Theorem is just the limit form of the derivative of  $x^n$  at  $x = a$ .

To Evaluate such limits:

Take the Power n.

Multiply it with  $a^{n-1}$ .

$$\text{Quest)} \lim_{x \rightarrow 1} \frac{x^{15} - 1}{x^{10} - 1}$$

$$\Rightarrow \lim_{x \rightarrow 1} \frac{x^{15} - 1^{15}}{x - 1} \times \frac{x - 1}{x^{10} - 1^{10}}$$

$$\Rightarrow \lim_{x \rightarrow 1} \frac{x^{15} - 1^{15}}{x - 1} \times \lim_{x \rightarrow 1} \frac{x - 1}{x^{10} - 1^{10}}$$

$$\Rightarrow 15(1)^{15-1} \times \frac{1}{10(1)^{10-1}}$$

$$\Rightarrow \frac{15}{10} \Rightarrow \left(\frac{3}{2}\right) \text{ Ans.}$$

## Continuity:

A function is said to be continuous (exist).

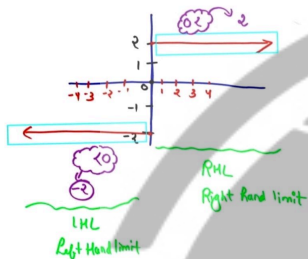
If  $LHL = RHL = f(a)$

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} f(x) = f(a)$$

$$\lim_{h \rightarrow 0} f(a-h) = \lim_{h \rightarrow 0} f(a+h) = f(a)$$

continuous.

$$LHL = RHL = f(a)$$



Question) Find  $\lim_{x \rightarrow 1} f(x)$  where  $f(x) = \begin{cases} x^2-1, & x \leq 1 \\ -x^2-1, & x > 1 \end{cases}$

$$= \begin{cases} x^2-1, & x \leq 1 \\ -x^2-1, & x > 1 \end{cases}$$

$$f(a) \\ x=1$$

$$\lim_{x \rightarrow 1} f(x)$$

$$\lim_{x \rightarrow 1} x^2-1$$

$$(1)^2-1$$

$$0$$

$$f(a^+) \\ RHL \ x > 1$$

$$\lim_{x \rightarrow 1^+} f(x)$$

$$\lim_{x \rightarrow 1^+} -x^2-1$$

$$-(1)^2-1$$

$$-1-1$$

$$-2$$

$$f(a^-) \\ LHL \ x < 1$$

$$\lim_{x \rightarrow 1^-} f(x)$$

$$\lim_{x \rightarrow 1^-} (x^2-1)$$

$$1^2-1$$

$$0$$

$$f(a) = f(a^-) \neq f(a^+)$$

∴ Hence, the given func<sup>n</sup> are not continuous ⇒ Limit doesn't exist.

Question) Find  $\lim_{x \rightarrow 0} f(x)$  and  $\lim_{x \rightarrow 0} f(x)$

$$\text{Where } f(x) = \begin{cases} 2x+3, & x \leq 0 \\ 3(x+1), & x > 0 \end{cases}$$

$$f(x) \\ x=0$$

$$f(x) \\ x>0 \text{ RHL}$$

$$f(x) \\ x<0 \text{ LHL}$$

$$\lim_{x \rightarrow 0} f(x)$$

$$\lim_{x \rightarrow 0^+} f(x)$$

$$\lim_{x \rightarrow 0^-} f(x)$$

$$\lim_{x \rightarrow 0} (2x+3)$$

$$\lim_{x \rightarrow 0^+} 3(x+1)$$

$$\lim_{x \rightarrow 0^-} 2x+3$$

$$\Rightarrow 2(0)+3$$

$$\Rightarrow 3(0+1)$$

$$\Rightarrow 2(0)+3$$

$$\Rightarrow 3$$

$$\Rightarrow 3$$

$$\Rightarrow 3$$

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^-} f(x) = 3$$

Ans.

## DERIVATIVE

What is a Derivative?

Basic Idea:

- A derivative tells us how a function is changing.
- It shows the rate of change of one quantity with respect to another.

$$\begin{array}{c} \text{Derivative } \frac{dy}{dx} \\ \downarrow \\ \text{Denote } \begin{cases} y' \\ f'(x) \end{cases} \end{array}$$

Example

$$y = x^3$$

$$\frac{dy}{dx} = 3x^2 \quad y' = 3x^2$$

$$i) \frac{d}{dx} (x^n) = nx^{n-1}$$

$$ii) \frac{d}{dx} (\sin x) = \cos x$$

$$iii) \frac{d}{dx} (\cos x) = -\sin x$$

$$iv) \frac{d}{dx} (\tan x) = \sec^2 x$$

$$v) \frac{d}{dx} (\cot x) = -\operatorname{cosec}^2 x$$

$$vi) \frac{d}{dx} (\sec x) = \sec x \tan x$$

$$vii) \frac{d}{dx} (\operatorname{cosec} x) = -\operatorname{cosec} x \cot x$$

$$viii) \frac{d}{dx} (a^x) = a^x \log_e a$$

$$1x > \frac{d}{dx}(e^x) = e^x$$

$$x > \frac{d}{dx}(\log_e x) = \frac{1}{x}$$

Product Rule:

$$\frac{d}{dx}(I \times II) = I \frac{d}{dx}(II) + II \frac{d}{dx}(I)$$

Quotient Rule:

$$\frac{d}{dx} \left[ \frac{I}{II} \right] = \frac{II \frac{d}{dx}(I) - I \frac{d}{dx}(II)}{(II)^2}$$

KBDT:

$$Q. x^4 + x^3 + 2 \rightarrow \text{Ans) } 4x^3 + 3x^2$$

$$Q. x^4 + 4 \rightarrow \text{Ans) } 2x$$

$$Q. x^3 - 27 \rightarrow \text{Ans) } 3x^2$$

$$Q. x^3 + 4x \rightarrow \text{Ans) } 3x^2 + 4$$

$$Q. f(x) = x^2 + 2$$

$$\rightarrow f'(x) = \frac{d}{dx}(x^2 + 2)$$

$$= \frac{d}{dx}(x^2) + \frac{d}{dx}(2)$$

$$= 2x + 0 = 2x \text{ Ans.}$$

Q. Find the derivative of following functions from  $\frac{x+1}{x-1}$ .

$$f(x) = \frac{x+1}{x-1} \quad \text{--- (I)}$$

$$f(x) = \frac{x+1}{x-1} \quad \text{--- (II)}$$

$$\frac{d}{dx} \left( \frac{I}{II} \right) = \frac{II \frac{d}{dx}(I) - I \frac{d}{dx}(II)}{(II)^2}$$

$$\rightarrow f'(x) = \frac{d}{dx} \left( \frac{x+1}{x-1} \right)$$

$$= \frac{(x-1) \frac{d}{dx}(x+1) - (x+1) \frac{d}{dx}(x-1)}{(x-1)^2}$$

$$= \frac{(x-1)(1+0) - (x+1)(1-0)}{(x-1)^2}$$

$$= \frac{(x-1) - (x+1)}{(x-1)^2}$$

$$\Rightarrow \frac{x-1-x-1}{(x-1)^2} \therefore f'(x) = \frac{-2}{(x-1)^2} \text{ Ans.}$$

## # First Order derivative Test

Suppose  $f$  is a real valued function and  $a$  is a point in its domain of definition.

The derivative of  $f$  at  $a$  is defined by  $\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$

Provided this limit exists. Derivative of  $f(x)$  at  $a$  is denoted by  $f'(a)$ . Observe that  $f'(a)$  quantifies the change in  $f(x)$  at  $a$  with respect to  $x$ .

Q. Find the derivative of  $\frac{x^n - a^n}{x - a}$  for some constant  $a$ .

$$\frac{d}{dx} \left( \frac{x^n - a^n}{x - a} \right) \quad \text{--- (I)}$$

$$\Rightarrow \frac{(x-a) \frac{d}{dx}(x^n - a^n) - (x^n - a^n) \frac{d}{dx}(x-a)}{(x-a)^2}$$

$$\Rightarrow \frac{(x-a)(nx^{n-1} - 0) - (x^n - a^n)(1-0)}{(x-a)^2}$$

$$\Rightarrow \frac{(x-a)(nx^{n-1}) - (x^n - a^n)(1-0)}{(x-a)^2}$$

$$\Rightarrow \frac{x \cdot nx^{n-1} - a \cdot nx^{n-1} - x^n + a^n}{(x-a)^2}$$

$$\Rightarrow \frac{nx^{n-1+1} - anx^{n-1} - x^n + a^n}{(x-a)^2}$$

$$\Rightarrow \frac{nx^n - anx^{n-1} - x^n + a^n}{(x-a)^2}$$

$$\frac{d}{dx}(x^n - a^n) \quad \left| \quad \frac{d}{dx} x^n - a^n \frac{d}{dx}(1) \right|$$

$$\Rightarrow \frac{d}{dx} x^n - \frac{d}{dx} a^n \Rightarrow nx^{n-1} - a^n(0)$$

$$\Rightarrow nx^{n-1} - 0$$

Product Rule:

$$\frac{d}{dx}(I \times II) = I \frac{d}{dx}(II) + II \frac{d}{dx}(I)$$

<sup>Imp</sup> Quest) Find the derivative of  $(5x^3 + 3x - 1)^4(x - 1)$

$$\Rightarrow \frac{d}{dx} \underbrace{(5x^3 + 3x - 1)}_I \underbrace{(x - 1)}_{II}$$

$$\Rightarrow (5x^3 + 3x - 1) \frac{d}{dx}(x - 1) + (x - 1) \frac{d}{dx}(5x^3 + 3x - 1)$$

$$\Rightarrow (5x^3 + 3x - 1)(1 - 0) + (x - 1)(15x^2 + 3)$$

$$\Rightarrow 5x^3 + 3x - 1 + (x - 1)(15x^2 + 3)$$

$$\Rightarrow 5x^3 + 3x - 1 + 15x^3 + 3x - 15x^2 - 3$$

$$\Rightarrow 20x^3 - 15x^2 + 6x - 4 \text{ Ans.}$$

Quest) Find the derivative of  $x^{-3}(5 + 3x)$ .

$$\frac{d}{dx}(x^{-3})(5 + 3x)$$

$$\Rightarrow x^{-3} \frac{d}{dx}(5 + 3x) + (5 + 3x) \frac{d}{dx}(x^{-3})$$

$$\Rightarrow (x^{-3})(0 + 3(1)) + (5 + 3x)(-3x^{-4})$$

$$\Rightarrow (x^{-3})(3) + (5 + 3x)(-3x^{-4})$$

$$\Rightarrow 3x^{-3} - 3x^{-4}(5 + 3x)$$

$$\Rightarrow 3x^{-3} - 15x^{-4} - 9x^{-3}$$

$$\Rightarrow -15x^{-4} - 6x^{-3} \text{ Ans.}$$

$$\text{Reference } \Rightarrow x^{-3} \frac{d}{dx}(5 + 3x) + (5 + 3x) \frac{d}{dx} x^{-3}$$

$$\Rightarrow (x^{-3}) \left[ \frac{d}{dx}(5) + \frac{d}{dx}(3x) \right] + "$$

$$\Rightarrow x^{-3} \left[ 5 \frac{d}{dx}(1) + 3 \frac{d}{dx}(x) \right] + (5 + 3x) \frac{d}{dx} x^{-3}$$

$$\Rightarrow x^{-3} [5(0) + 3(1)] + (5 + 3x)(-3x^{-4})$$

Quest) Find the derivative of  $\cos x$  from first principle.

$$\text{Formula: } \frac{f(x+h) - f(x)}{h}$$

$$\lim_{h \rightarrow 0} \frac{\cos x \cdot \cosh - \sin x \sinh - \cos x}{h}$$

$$\lim_{h \rightarrow 0} \frac{\cos x \cdot \cosh - \cos x - \sin x \sinh}{h}$$

$$\lim_{h \rightarrow 0} \frac{\cos x (\cosh - 1) - \sin x \sinh}{h}$$

$$\lim_{h \rightarrow 0} \frac{\cos x (\cosh - 1)}{h} - \frac{\sin x \sinh}{h}$$

$$\lim_{h \rightarrow 0} -\frac{\cos x (1 - \cosh)}{h} - \frac{\sin x \sinh}{h}$$

$$\Rightarrow -\cos x \lim_{h \rightarrow 0} \frac{1 - \cosh}{h} - \sin x \lim_{h \rightarrow 0} \frac{\sinh}{h}$$

$$\Rightarrow -\cos x(0) - \sin x(1)$$

$$\Rightarrow (-\sin x) \text{ Ans.}$$

| Differentiation Rules |                                                                                       |
|-----------------------|---------------------------------------------------------------------------------------|
| Constant Rule         | $\frac{d}{dx}[c] = 0$                                                                 |
| Power Rule            | $\frac{d}{dx}x^n = nx^{n-1}$                                                          |
| Product Rule          | $\frac{d}{dx}[f(x)g(x)] = f'(x)g(x) + f(x)g'(x)$                                      |
| Quotient Rule         | $\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{g(x)f'(x) - f(x)g'(x)}{[g(x)]^2}$ |
| Chain Rule            | $\frac{d}{dx}[f(g(x))] = f'(g(x))g'(x)$                                               |